Deliverable Templates and Checklists

Editable Structures for Team and Project Evidence

Module	Software Architecture and Process Models
Student Level	Level 2
Assignment Duration	7 weeks
Document	07 of 07
Purpose	Provide concise academic templates, not commercial handover documents
Excluded	Warranty, production credentials handover, and commercial sign-off

1. How to Use These Templates

- Adapt each template to the assigned project and team domain; do not submit blank sections.
- Keep one authoritative version of each project-level document in the shared project-docs repository.
- Link Jira issues, pull requests, diagrams, API specifications, and test evidence wherever possible.
- Use version history and clear ownership. Update documents as the system changes.
- Concise, accurate evidence is preferred to lengthy generic text.

2. Project Proposal and Charter Template

Section	Required Content
Project title and summary	Problem, intended users, proposed solution, and expected value in 150-250 words.
Objectives	Specific and measurable project objectives for the seven-week period.
Scope	In-scope capabilities, out-of-scope items, assumptions, and constraints.
Stakeholders	Primary users, project teams, teaching team, and relevant service roles.
Team allocation	Seven members, primary/secondary roles, and contact method.
Technology and architecture direction	React/TypeScript/Redux, Spring Boot, MySQL, JWT, Docker, API Gateway, GitHub, Jira, and deployment approach.
Milestones	Week-by-week milestones and chosen sprint cadence.
Risks	Initial technical, integration, schedule, scope, data, security, and collaboration risks.
Success criteria	Working workflows, quality evidence, deployment, documentation, and final demonstration conditions.

3. Project Plan and Milestone Template

Week / Sprint	Goal	Planned Epics / Stories	Dependencies	Owner(s)	Evidence / Exit Condition
Week 1					
Week 2					
Week 3					
Week 4					
Week 5					
Week 6					
Week 7					

4. Risk Register Template

ID	Risk	Probability	Impact	Response / Mitigation	Owner	Status / Review Date
R-01	Example: Dependent API is delivered late.	Medium	High	Agree contract early; provide mock; track dependency; integration checkpoint.	Backend Developer	Open / date

5. Software Requirements Specification Template

1. Introduction: purpose, product scope, intended audience, definitions, references.
2. Overall description: users, operating environment, assumptions, dependencies, constraints.
3. System context: external actors, services, and project boundaries.
4. Functional requirements: uniquely identified and grouped by business capability.
5. User stories and acceptance criteria: linked to Jira issues.
6. Business rules: validations, eligibility, status transitions, conflicts, and authorization rules.
7. External interface requirements: frontend, REST APIs, authentication, and third-party or simulated interfaces.
8. Data requirements: owned entities, relationships, retention assumptions, synthetic data, and privacy rules.
9. Non-functional requirements: security, usability, reliability, maintainability, performance, logging, and deployment.
10. Traceability matrix: requirement to story, design component, implementation, and test evidence.
11. Scope exclusions and known limitations.
12. Version history and approvals.

6. Functional Requirement Template

Field	Example / Guidance
Requirement ID	AMS-BILL-FR-01
Title	Generate monthly invoice
Statement	The system shall allow an authorized finance officer to generate an invoice for an eligible unit and billing period.
Preconditions	User is authenticated and authorized; unit exists; charge rules are active.
Inputs	Unit, billing period, applicable charges.
Main outcome	Invoice and line items are stored with Issued status and displayed to the eligible user.
Exceptions	Duplicate period, invalid unit, no applicable charges, unavailable dependency.
Priority	Must / Should / Could
Linked Jira story	AMS-G3-24
Linked tests	TC-BILL-01, API-BILL-04, E2E-02

7. User Story Template

Field	Content
Story	As a [role], I want [capability], so that [benefit].
Acceptance criteria	Use clear testable bullets or Given-When-Then statements.
Priority and estimate	Priority plus story points or selected estimation method.
Dependencies	External Jira issue, API contract, design, data, or team dependency.
Definition of Done additions	Any story-specific documentation, test, migration, or release requirement.

8. Requirements Traceability Matrix Template

Requirement ID	Jira Story	Service / UI Module	API / Design Reference	Test Case(s)	Status

9. Architecture Document Template

1. Architecture goals and quality attributes.
2. System context diagram and external actors.
3. Microservice architecture diagram with service owners and responsibilities.
4. Service boundaries, key APIs, and inter-service communication.
5. Data ownership and ERD/data model for each service.
6. API Gateway and authentication/authorization flow.
7. Shared frontend architecture, feature modules, Redux state ownership, routing, and common components.
8. Deployment diagram including Docker containers, databases, ports, environment configuration, and cloud components.
9. Error handling, timeouts, logging, health checks, and failure scenarios.
10. Security and threat considerations.
11. Architecture decisions and trade-offs.
12. Known limitations and planned improvements.

10. Architecture Decision Record Template

Field	Required Content
ADR ID and title	ADR-003: Separate reservation and facility services
Status	Proposed / Accepted / Superseded
Context	Problem, constraints, and forces that require a decision.
Decision	The chosen approach stated clearly.
Alternatives considered	At least one credible alternative and why it was not selected.

Field	Required Content
Consequences	Benefits, costs, risks, migration, and impact on other teams.
Date and participants	Decision date and teams/roles involved.
Related evidence	Diagram, pull request, Jira change, API contract, or test.

11. API Design Specification Template

Field	Example / Guidance
Service and version	reservation-service / v1
Endpoint	POST /api/v1/reservations
Purpose and owner	Create a reservation request; Group 6.
Authentication and roles	Bearer JWT; Student, AcademicStaff, AdministrativeStaff.
Request schema	Fields, types, required constraints, and example JSON.
Response schema	Success response and example JSON.
Validation rules	Eligibility, time period, conflict, capacity, and status rules.
Status codes	201, 400, 401, 403, 409, 503 with meanings.
Dependencies and failure behavior	Identity validation, facility availability, timeout, unavailable-service response.
Version/change notes	Backward-compatible or breaking change details.
Tests	Postman request and automated integration test references.

12. UI/UX Deliverable Checklist

[ ] User flow or journey for each major workflow.
[ ] Low-fidelity wireframes for key screens before implementation.
[ ] High-fidelity designs or implemented design references for key screens.
[ ] Shared navigation, page layout, typography, components, form conventions, and status styles.
[ ] Role-aware dashboard and navigation behavior.
[ ] Responsive behavior for common desktop and mobile-width browser views.
[ ] Accessible labels, keyboard use where practical, readable contrast, and useful validation messages.
[ ] Empty, loading, success, warning, and error states.
[ ] Design-to-implementation review evidence.

13. Test Plan Template

Section	Required Content
Objectives and scope	What will be verified and what is excluded.
Test levels	Unit, API, integration, system, security, performance observation, and simulated UAT.
Test environment	Versions, services, databases, Docker, URLs, and synthetic test data.
Roles and responsibilities	Test design, execution, defect triage, fix, retest, and approval responsibilities.
Entry criteria	Build available, required services running, test data ready, stories ready for QA.

Section	Required Content
Exit criteria	Critical tests pass, critical defects closed, accepted residual limitations documented.
Schedule	Planned test design, execution, integration checkpoints, UAT, and final retest.
Risks and contingencies	Environment failure, dependent service delay, data reset, or deployment outage.
Reporting	Where cases, results, evidence, defects, and summary metrics are stored.

14. Test Case Template

Field	Content
Test Case ID	
Requirement / Story	
Title and objective	
Preconditions and test data	
Steps	1. ... 2. ... 3. ...
Expected result	
Actual result	
Status	Pass / Fail / Blocked / Not Run
Evidence	Screenshot, log, API response, CI result, or video reference.
Defect ID	
Tester and date	

15. Defect Log Template

Defect ID	Summary	Severity	Environment	Owner	Status	Fix / PR	Retest Result

16. Test Execution Report Template

1. Release/build tested and environment.
2. Scope and test period.
3. Summary counts: planned, run, passed, failed, blocked, and not run.
4. Results by requirement, service, or workflow.
5. Critical defects and resolution status.
6. Security checks and findings.

7. Performance observations and limitations.
8. Simulated UAT participants, scenarios, findings, and accepted changes.
9. Residual risks and recommendation for final release.
10. Evidence links.

17. Progress Report Template

Section	Questions to Answer
Reporting period	What dates and sprint/iteration does this cover?
Completed	Which stories, tasks, designs, tests, integrations, and documents were completed?
In progress	What is currently being worked on and by whom?
Blocked / delayed	What is blocked, why, dependency owner, impact, and next action?
Quality status	Build, CI, defects, tests, integration, and documentation status.
Risks and changes	New or changed risks, scope decisions, or API changes.
Next period	Planned outcomes and cross-team commitments.
Evidence	Jira dashboard, pull requests, deployment, tests, or meeting notes.

18. Change Request Template

Field	Content
Change ID and date	
Requested by	
Current requirement / design	
Proposed change and reason	
Scope impact	
Architecture / API / data impact	
Schedule and effort impact	
Testing and documentation impact	
Affected teams and Jira issues	
Decision	Approved / Rejected / Deferred
Decision rationale and approvers	

19. Sprint Review Template

- Sprint goal and planned stories.
- Completed and not-completed work with Jira evidence.
- Demonstrated increment and test evidence.
- Feedback received and resulting backlog changes.
- Integration status and dependency commitments.
- Known defects, risks, and next sprint priorities.

20. Sprint Retrospective Template

What helped us?	What slowed us down?	What will we change next sprint?	Owner and due date

21. Release Notes Template

1. Release name, date, and Git tag.
2. Compatible repositories and versions.
3. Implemented features and completed workflows.
4. Fixed defects.
5. Database migrations or setup changes.
6. Configuration changes without secrets.
7. Known limitations and residual defects.
8. Deployment or rollback notes.
9. Test summary and release recommendation.

22. Deployment Guide Template

1. Prerequisites and supported versions.
2. Repository checkout and release tags.
3. Environment variables and configuration examples.
4. Database creation or migrations.
5. Docker image build or pull steps.
6. Docker Compose local startup order and health verification.
7. Cloud deployment steps and required platform services.
8. API Gateway and frontend configuration.
9. Smoke-test steps and expected results.
10. Troubleshooting and rollback guidance.
11. Known limitations.

23. User and Administrator Guide Template

Guide	Minimum Content
User guide	Purpose, supported roles, login, navigation, major workflows, validation messages, status meanings, and common problems.
Administrator guide	User/role management, master data, approvals, operational monitoring, reports, configuration boundaries, and recovery guidance.
Visual evidence	Use concise screenshots or annotated steps from the final release; do not include real personal data or credentials.

24. Project Closure Report Template

1. Project summary and final scope delivered.
2. Objectives achieved and objectives not fully achieved.
3. Final architecture and deployed release reference.
4. Requirements and workflow completion summary.
5. Quality summary, open defects, and known limitations.
6. Schedule, scope, and risk performance.
7. Cross-team collaboration outcomes and integration challenges.
8. Lessons learned and recommended future work.
9. Repository, Jira, documentation, and deployment links.

25. Individual Contribution Statement Template

Field	Content
Student name and number	
Team and primary/secondary roles	
Main requirements/design contributions	Jira or document links.
Main implementation contributions	Repositories, commits, and pull requests.
Code reviews and integration work	Pull requests reviewed, defects fixed, integration tasks.
Testing / QA / deployment contributions	Test cases, automation, defects, CI, Docker, or release work.
Documentation / coordination contributions	Reports, API docs, guides, risk/change management, liaison work.
Most important learning	100-150 words.
Declaration	I confirm that this statement accurately represents my contribution and that sources and assistance have been acknowledged.

26. Final Project Submission Checklist

[ ] Project proposal/charter and role allocation are final.
[ ] Jira projects contain final sprint, backlog, bug, change, and progress evidence.
[ ] SRS, business rules, user stories, and traceability are current.
[ ] Architecture diagrams, ERDs, API specifications, and ADRs match the delivered system.
[ ] Shared frontend, API Gateway, microservices, databases, and cross-team workflows integrate successfully.
[ ] All repositories have README files, release tags, configuration examples, and no committed secrets.
[ ] GitHub Actions build and test required repositories.
[ ] Swagger/OpenAPI and Postman assets are complete.
[ ] Test plan, cases, execution report, security review, performance observations, UAT simulation, and defect log are complete.
[ ] Docker Compose local setup works from the documented steps.
[ ] Cloud or approved environment deployment is accessible and smoke-tested.
[ ] User guide, administrator guide, deployment guide, dependency list, and release notes are complete.
[ ] Final presentation and demonstration data are prepared.
[ ] Project closure report, retrospective, and lessons learned are complete.
[ ] Each student submitted an individual contribution statement with verifiable links.